

LINKED LIST

PROBLEM 9

QUESTION:

DS	Session	Linked List	Question Information	Level 1	Challenge 39
Question description sanam's Dream came true after he got an Appointment order from Google.Simon's family was very happy of his achievement. The company mentioned Basic Salary, DA, HRA with some other benefits. But not highlighted the Gross salary in the order. sanam's father wanted to know the Gross salary of his son. sanam try to his gross salary from HR department. they informed that you have to get pass grade in first month entry test. the entry test has 5 questions. one of the question was, Split a circular linked list in two halves. you have to split the circular Linked List with the same size of Divisions.Maybe if circular Linked List is odd, you have to change the number of node, it is even . Can you help sanam? Function Description First count the number of node in Circular Linked List. Second, you have to make the List even. Third, you need to make the List half. the front is the same size like the rear. Finally, I have to make two circular Linked List. Constraints $0 < n < 10$ Input Format: The First line represents the number of input in the circular linked list elements Output Format: First Line indicates the complete linked list second line indicates the odd list					

third line indicates the even list

Refer sample test cases.

Logical Test Cases

Test Case 1

INPUT (STDIN)

6

EXPECTED OUTPUT

Complete linked_list:
[h]==1=>2=>3=>4=>5=>6=>[h]
Odd:
[h]==1=>3=>5=>[h]
Even:
[h]==2=>4=>6=>[h]

Test Case 2

INPUT (STDIN)

15

EXPECTED OUTPUT

Complete linked_list:
[h]==1=>2=>3=>4=>5=>6=>7=>8=>9=>10=>11=>12=>13=>14=>15=>[h]
Odd:
[h]==1=>3=>5=>7=>9=>11=>13=>15=>[h]
Even:
[h]==2=>4=>6=>8=>10=>12=>14=>[h]

Mandatory Test Cases

Test Case 1

KEYWORD

void insert(int data)

Test Case 2

KEYWORD

struct n *next;

Test Case 3

KEYWORD

void display(struct n *h)

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

4

Test Case 2

TOKEN COUNT

620

Test Case 3

NLOC

115

SOURCE CODE:

```
#include <stdio.h>
#include <stdlib.h>

struct n {
    int data;
    struct n *next;
};

struct n *head = NULL;
struct n *tail = NULL;

void insert(int data) {
    struct n *new_node = (struct n*)malloc(sizeof(struct n));
    new_node->data = data;
    new_node->next = NULL;
    if (head == NULL) {
        head = new_node;
        tail = new_node;
        new_node->next = head;
    } else {
        tail->next = new_node;
        tail = new_node;
        tail->next = head;
    }
}

void display(struct n *h) {
    if (h == NULL) return;
    struct n *temp = h;
    printf("[h]=>");
    do {
        printf("%d=>", temp->data);
        temp = temp->next;
    } while (temp != h);
    printf("[h]\n");
}

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;
    for (int i = 1; i <= n; i++) {
        insert(i);
    }
    printf("Complete linked_list:\n");
    display(head);

    struct n *odd_head = NULL, *odd_tail = NULL;
    struct n *even_head = NULL, *even_tail = NULL;
    struct n *curr = head;
    int count = 1;
    do {
        struct n *new_node = (struct n*)malloc(sizeof(struct n));
        new_node->data = curr->data;
        new_node->next = NULL;
        if (count % 2 != 0) {
            if (odd_head == NULL) {
                odd_head = new_node;
                odd_tail = new_node;
                new_node->next = odd_head;
            } else {
                odd_tail->next = new_node;
                odd_tail = new_node;
                odd_tail->next = odd_head;
            }
        } else {
            if (even_head == NULL) {
                even_head = new_node;
                even_tail = new_node;
                new_node->next = even_head;
            } else {
                even_tail->next = new_node;
                even_tail = new_node;
                even_tail->next = even_head;
            }
        }
        count++;
        curr = curr->next;
    } while (curr != head);

    printf("Odd:\n");
    display(odd_head);
    printf("Even:\n");
    display(even_head);

    return 0;
}
```

OUTPUT:

```
6
Complete linked_list:
[h]=>1=>2=>3=>4=>5=>6=>[h]
Odd:
[h]=>1=>3=>5=>[h]
Even:
[h]=>2=>4=>6=>[h]
```

```
15
Complete linked_list:
[h]=>1=>2=>3=>4=>5=>6=>7=>8=>9=>10=>11=>12=>13=>14=>15=>[h]
Odd:
[h]=>1=>3=>5=>7=>9=>11=>13=>15=>[h]
Even:
[h]=>2=>4=>6=>8=>10=>12=>14=>[h]
```